



Introduction and Background

How does pancreatic cancer form?

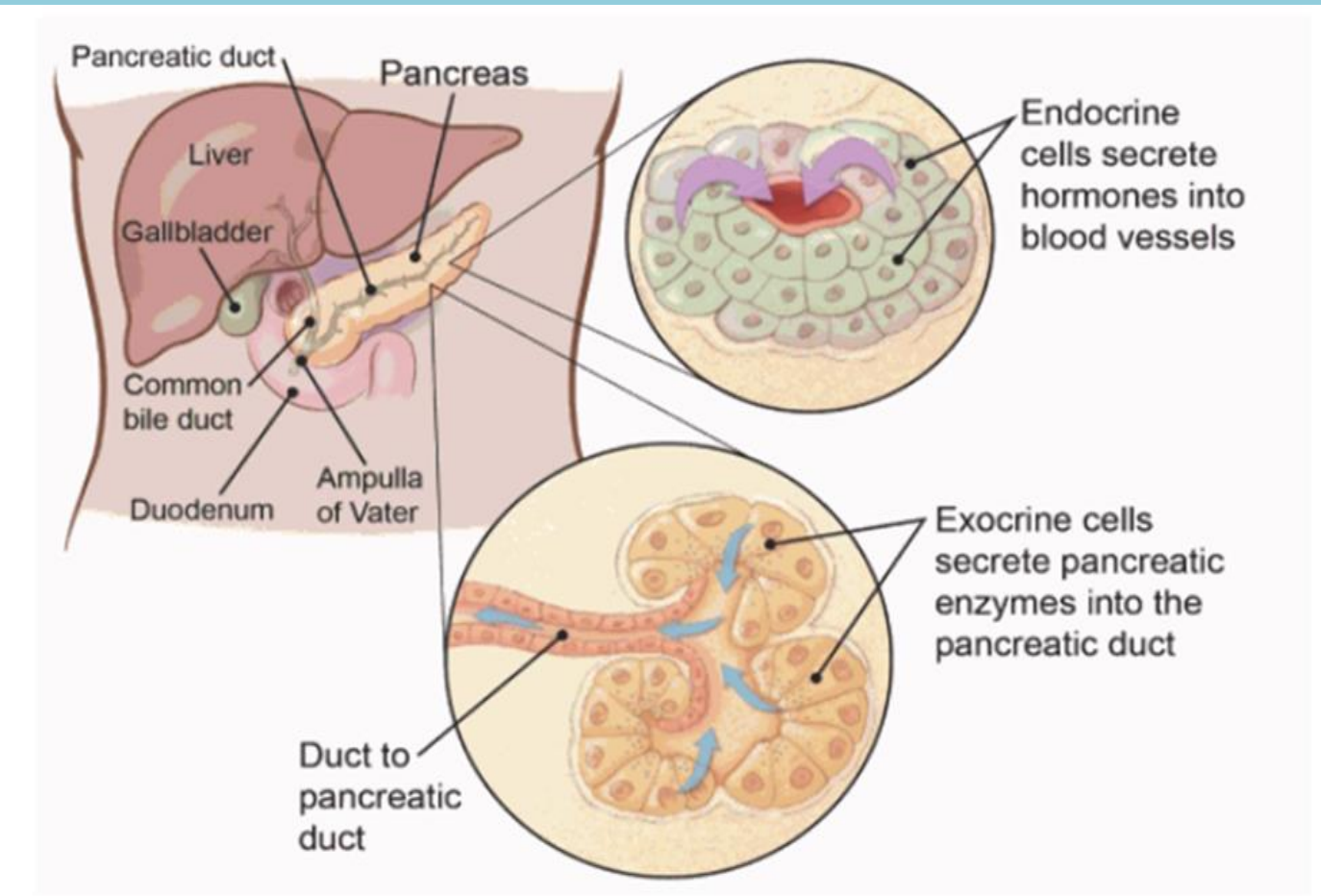


Figure 1: Pictorial representation of Pancreas System

- The pancreas is mostly made up of exocrine cells, which produce the exocrine glands and ducts seen in it
- PDAC, the most prevalent kind of pancreatic cancer, develops when exocrine cells in the pancreas begin to grow uncontrollably
- These cells also continue living after normal cells would usually die
- The accumulating cells then go on to form a tumor

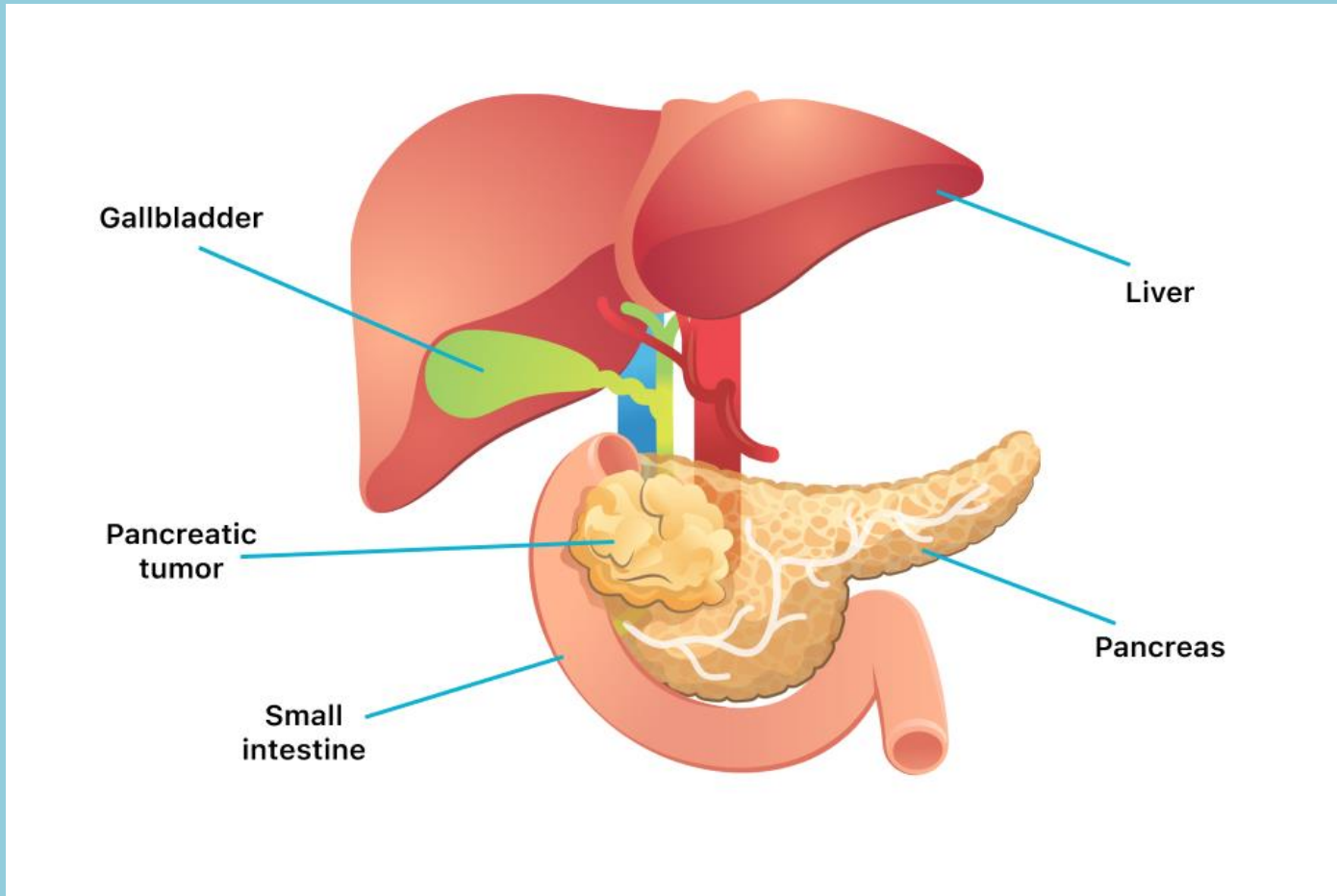


Figure 2: Pictorial representation of a Pancreas tumor

Current Statistics

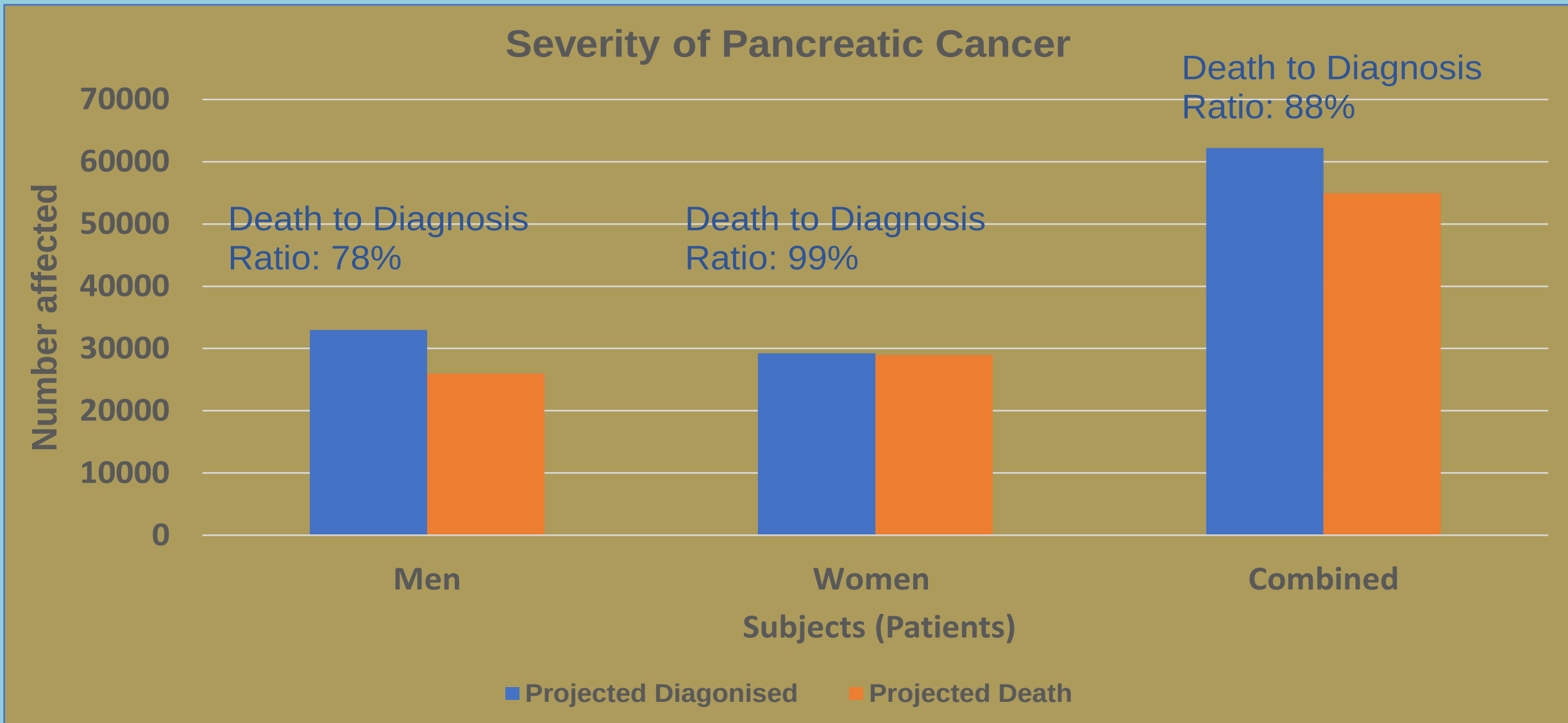


Figure 3(a). The severity of pancreatic cancer in men and women- USA based

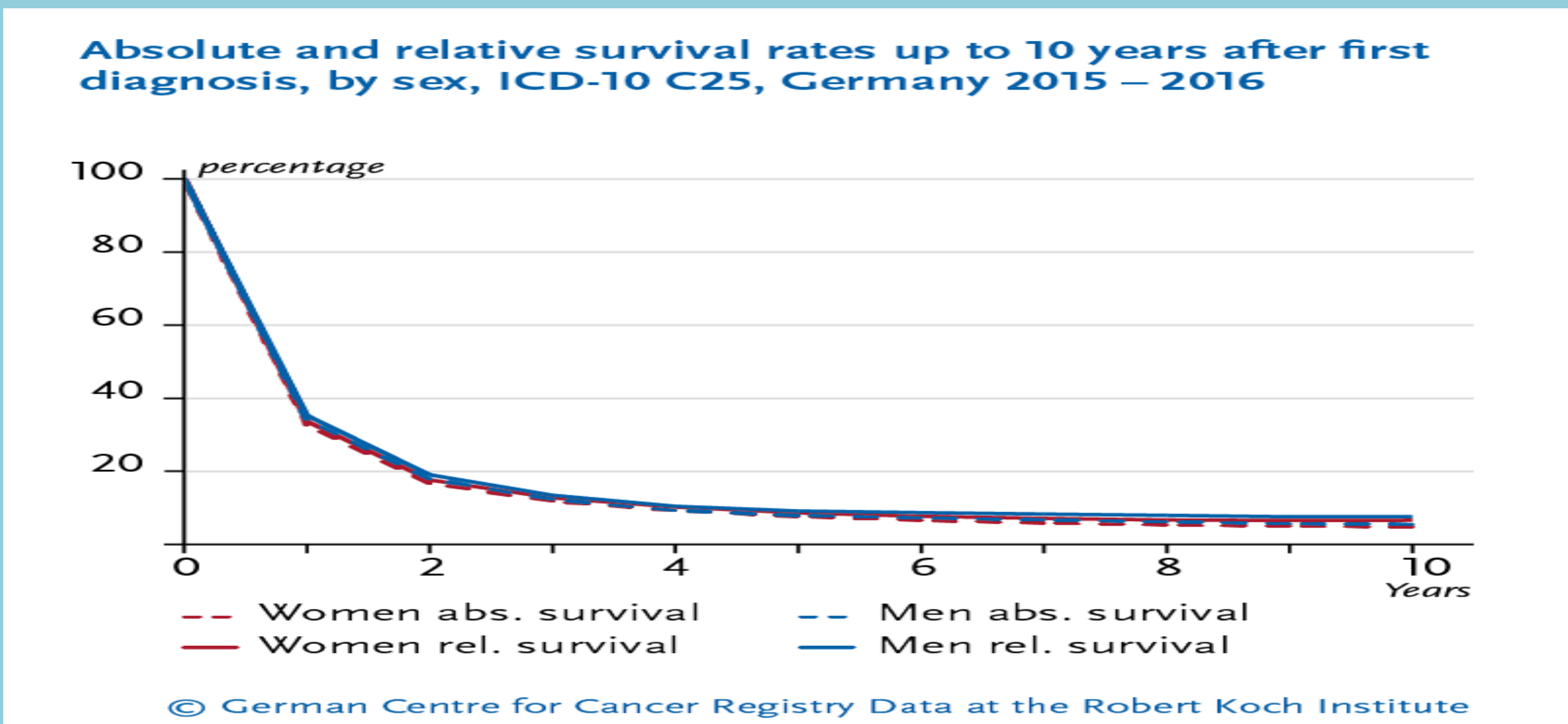


Figure 3(b). The severity of pancreatic cancer- Germany based

- Pancreatic cancer is projected to surpass breast and colorectal cancers to become the second leading cause of cancer-related death by 2030 after lung cancer
- Although pancreatic cancer only accounts for 3% of all cancers in the USA, it has a disproportionately high death rate (Fig 1)
- One of the most prevalent cancers where treatment planning through application of technology would benefit from early detection is pancreatic cancer
- Current diagnostic has significant limitation in early detection of PDAC
- Early detection of cancer is critical for successful treatment and improved survival rates, thus currently one of the most researched topics in the medical field

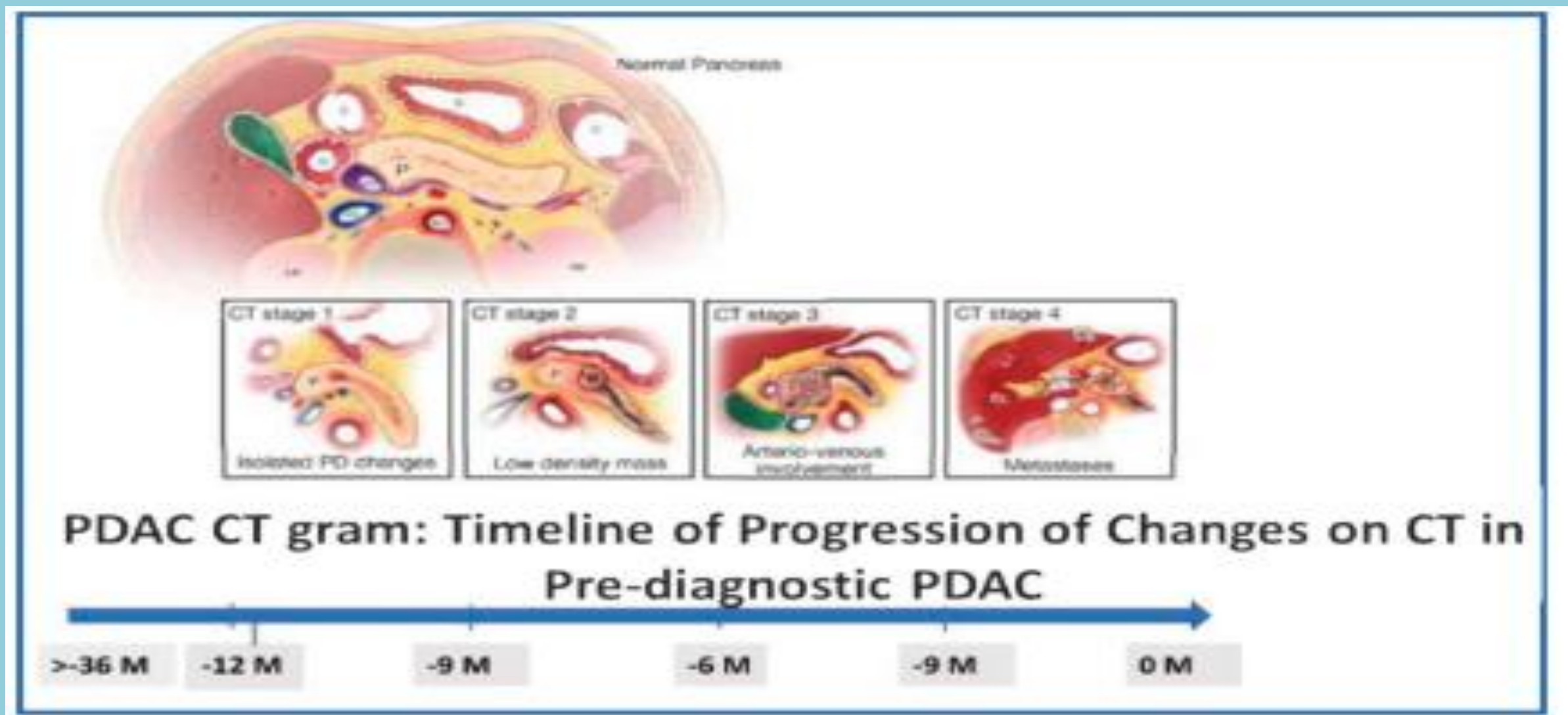


Figure 4: Barbara Kenner Et Al. Artificial Intelligence and Early Detection of Pancreatic Cancer, 2020 Summative Review Conference Report

Need for Early Detection

- There are no proven biomarkers available to diagnose early-stage PDAC
- Endoscopic ultrasonography and CT guided biopsies are the "gold standard" for pancreatic cancer diagnosis, however, they are ineffective in early-stage detection
- CT scans have a limited role in finding PDAC beyond 12 months from diagnosis (true false-negative).
- Early detection is they key for successful treatment of PDAC.
- The role of artificial intelligence in radiomics to discern changes not visible to the human eye could be viable when CT scans are "normal" to the human eye and could lead to life altering decisions
- This early detection does not necessarily mean prevention, but it can lead to early treatment and improved outcomes, saves millions of life. Technological advancement in early detection has demonstrated huge success in medical field for some types of cancer

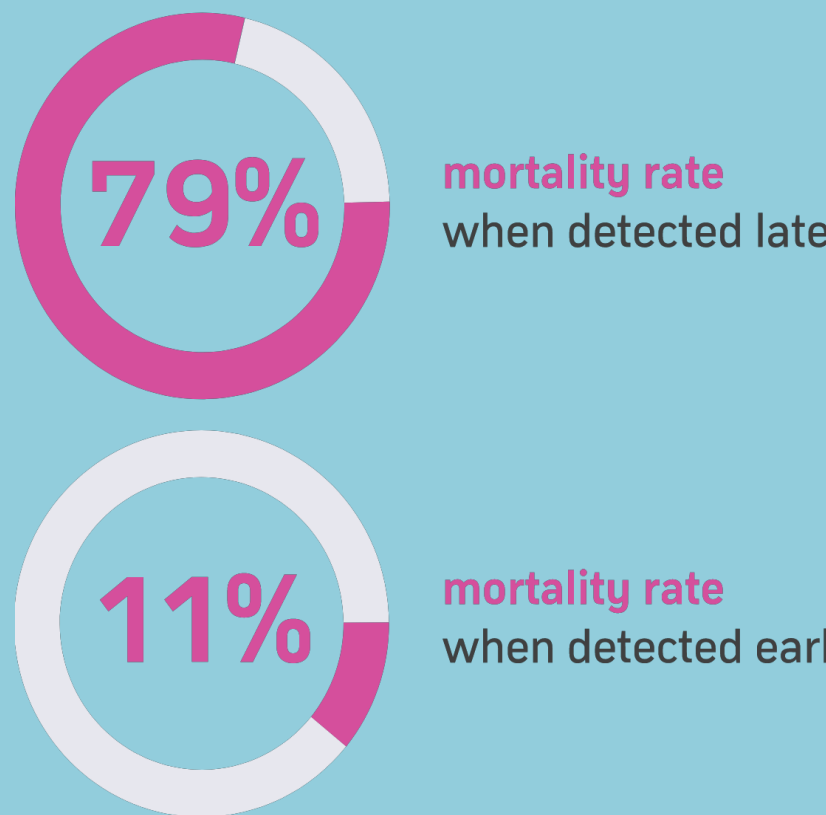


Table 1: Type of current diagnosis of pancreatic cancer and their limitations

Type of Diagnosis	Current limitations
Biomarkers	. An overall sensitivity in the range of 25% to 50% in early-stage disease identifying credentialed biomarkers for early detection of PDAC remains an area of great unmet need. . Is not always reliable for detecting early pancreatic cancer in general and performs poorly in detecting early-stage pancreatic cancer.
Endoscopic ultrasonography	. Mainly used to dragonize metastatic disease. . The potential for early detection of asymptomatic pancreatic neoplasms in high-risk individuals (HRI) using an endoscopic approach was first reported in 1999 but didn't progress much. . Invasive procedure which comes with risks such as bleeding or infection . Mainly used to dragonize metastatic disease.
Computed tomography (CT) guided biopsy	. The sensitivity of CT scan findings suspicious for PDAC at ~18 months was only 15%. Even 6 months before diagnosis, it was only ~50%. At 3 months, it was 85%. The patients were also exposed to radiation repeatedly. . Poor at diagnosing early stages of pancreatic cancer and also an invasive procedure which can cause bleeding or infections

Key Question

Can Medical Image Analysis by Deep Learning make a breakthrough in Early Detection of PDAC?

Goal

My goal is to integrate my AI algorithm into the diagnosis process as a “second reader”, that can improve diagnostic confidence and potentially increase the detection of early subtle pancreatic cancer that can be missed by a radiologist



Hypothesis

The detection of pancreatic cancer at early stages can be accomplished using a DL-based model architecture that can outperform early detection rates of pancreatic cancer made by human experts. Long-term, the treatment in pancreatic cancer will significantly improve with early detection by these models, due to the fact that DL models are being used extensively and successfully in many fields related to cancer, so there is a high possibility that it can also improve pancreatic cancer treatment